Hamza Benarfa

Full-Stack Software Engineer · Backend, Cloud & DevOps

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SUMMARY

Full-stack software engineer with 4+ years of experience building and running production web products end to end: architecture, backend, frontend, cloud infrastructure, and deployment. Owned DevOps at two startups, worked through architecture decisions directly with a CTO, mentored engineers, and built LLM features into real product workflows. Main stack: TypeScript, Next.js, NestJS, Python/FastAPI, PostgreSQL, AWS/Azure, Terraform, and Docker. Works best with full ownership of a system and a problem that is not yet fully defined.

EXPERIENCE

Robosim.AI — On-site

Software Engineer — Jul 2026 – Sep 2026

No-code, AI-powered robot simulation platform: describe a scene in plain English, simulate it in 3D, run it on real robots.

- Sole full-stack engineer on the web platform (Next.js, FastAPI, PostgreSQL), owning architecture, backend, frontend, infrastructure, CI/CD, and production releases.
- Designed the robotics App Store: prompts, imported projects, and store apps are converted to a robot-neutral representation, then pass compatibility checks, sandboxed build and simulation gates, and approval before fleet deployment. Adapter boundaries keep robot-specific code out of the core product.
- Built the interactive 3D warehouse and simulation editor with Next.js and React Three Fiber: floor plans, environments, robots, and applications.
- Built AI features: a warehouse-design agent, multi-provider LLM integration, and voice commands, each with deterministic validation and safety checks.
- Built multi-tenant authentication, billing, and long-running file-processing jobs, plus a SuperAdmin console for companies, users, subscriptions, and usage, with audited account impersonation.
- Automated AWS infrastructure with Terraform and Docker; set up GitLab CI/CD with security scanning and rollbacks; reduced cloud spend by right-sizing resources.
- Compared three migration routes for the monolith (full rework, clean-up in place, phased extraction) and presented the trade-offs to the CTO; phased extraction became the migration plan.
- Led the technical side of a small team: mentored a robotics engineer and an AI engineer, integrated their work into the platform, and reviewed code through GitLab merge requests. Kept fast delivery stable with TDD, integration and E2E tests, and CI quality gates.

D-Talk — Remote

Full-Stack Developer — Nov 2024 – Oct 2026

Fashion-tech platform where brands, designers, and fashion professionals build a brand step by step, from design and customization to orders, production, and delivery.

- Started on the frontend (role-based dashboards, browser-based product customizer), moved to backend APIs and services, then took over all infrastructure.
- Built the real-time creator and customization editor with Next.js, KonvaJS, and Zustand on a reusable component architecture.
- Owned DevOps end to end: production deployments, CI/CD, servers, monitoring, and Docker, on Azure infrastructure managed with Terraform.
- Moved the data layer to PostgreSQL with PgBouncer connection pooling to support production growth.
- Built custom B2B products for fashion and textile clients, including an AI-assisted fabric recommendation service in NestJS.
- Wrote the product spec for a 3D garment and AI visualization module, phased from fidelity validation to an AI photo and video render pipeline.

Freelance & Contract Remote
Full-Stack Developer Jun 2022 – Present

- Build web products for clients from first requirements to production, then maintain and extend them; work directly with clients on scope, architecture, and trade-offs, with no project manager in between.
- Take MVPs to production with TypeScript, Next.js, Node.js, Prisma, and PostgreSQL.
- **Tactix Solutions** (contract, Jun 2025 – Jul 2025): built and shipped a production e-commerce platform with React and NestJS on a modular architecture, with an accessible UI in Tailwind CSS and shadcn/ui.

NGB Professional Remote
Full-Stack Development Trainer Aug 2024 – Oct 2024

- Taught hands-on full-stack development with Node.js and React: Clean Architecture, Git Flow, project structure, and Docker deployment.
- Guided learners from first idea to deployed, portfolio-ready applications.

SELECTED PROJECTS

BeadCraft Studio · Founder craft.benarfa.com
Bracelet configurator for independent jewelry sellers: customers design their own piece before buying. Started as a white-label store; now being packaged as a Shopify app. Owns product, engineering, pricing, and sales.

Menu-qr.tn menu-qr.tn
Digital menu platform for Tunisian restaurants: QR-code menus with menu and order management.

EDUCATION

Higher Institute of Technological Studies of Béja (ISET Béja) Tunisia
Bachelor's Degree, Web & Mobile Development

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Python, SQL
Frontend & 3D	Next.js, React, React Three Fiber, Three.js, KonvaJS, Zustand, TanStack Query, Tailwind CSS, shadcn/ui
Backend & Data	NestJS, Node.js, FastAPI, Pydantic, SQLAlchemy, Prisma, REST, WebSockets, PostgreSQL, PgBouncer
Cloud & DevOps	AWS, Azure, Terraform, Docker, GitLab CI/CD, GitHub Actions, monitoring
AI	LLM integration, agent workflows, multi-provider LLM APIs, voice commands, streaming UX, recommendation systems
Practices	System design, TDD, Clean Architecture, integration and E2E testing (Playwright), code review, Git Flow, mentoring
Languages	Arabic (native), English (fluent), French (proficient)